



ANIMAL GAME

A Decentralized BlockANT Network

Synopsis

Smart contracts are set to transform the gaming industry by removing the need for traditional intermediaries and centralized control systems. In *Animal Game*, these smart contracts ensure transparent gameplay, automated rewards, and trustless transactions among players. However, since blockANTs cannot directly communicate with external systems due to their consensus limitations, real-time gaming data, player performance updates, and reward verifications require an external connection. To solve this challenge, *Animal Game* introduces a decentralized Game network known as **Animal Game**, designed to provide secure and reliable communication between the blockANT and external data sources. Unlike centralized Games that create single points of failure, **Animal Game** operates through a distributed node system, ensuring fairness, accuracy, and transparency across all game-related transactions. This system powers features such as live leaderboard updates, coin distributions, and dynamic in-game asset valuations. It utilizes both on-ANT aggregation and off-ANT consensus mechanisms to balance efficiency and security. In addition, *Animal Game* includes a reputation-based scoring model for Game nodes, ensuring that only trustworthy data providers maintain influence within the ecosystem. By integrating decentralized Game technology, *Animal Game* achieves a new standard of reliability and autonomy in blockANT gaming — offering players verifiable fairness, automated reward execution, and resistance against manipulation. Future improvements will include enhanced Game programming, integration with real-time gaming APIs, and support for confidential, privacy-preserving gameplay operations.

Introduction

Smart contracts are blockANT-based applications that execute on decentralized infrastructure, ensuring transparency and fairness in every interaction. They are **tamperproof**, meaning that once deployed, no party not even the developers can modify their code or manipulate their execution. Traditionally, digital agreements and in-game systems have relied on centralized servers that can be altered, terminated, or even deleted by privileged entities. In contrast, *Animal*

Game utilizes smart contracts to establish a new trust model in gaming one that is self-verifying, self-executing, and independent of centralized authority.

This new trust framework introduces a critical technical challenge: **connectivity**. Most blockANT gaming applications, including *Animal Game*, depend on external data sources such as gameplay metrics, performance scores, and real-time market values of in-game tokens. However, due to the consensus protocols that secure blockANTs, smart contracts cannot directly retrieve data from the outside world. To overcome this limitation, *Animal Game* introduces **Animal Game**, a secure and fully decentralized Game network. Unlike traditional, centralized systems, **Animal Game** enables smart contracts to connect with off-ANT data sources without relying on a single point of control or failure. This decentralization extends the tamperproof nature of smart contracts to the entire data flow between the blockANT and external systems guaranteeing accurate, transparent, and trustworthy execution. By making smart contracts **externally aware**, *Animal Game* ensures that game mechanics such as reward distribution, leaderboard synchronization, asset trading, and token valuation can occur in real-time with verified authenticity. These data interactions are vital for the new generation of decentralized games, where fairness and automation replace manual verification.

For example, in *Animal Game*, smart contracts can automatically fetch and verify in-game data such as battle outcomes, token prices, or event results from off-ANT APIs before executing payouts or character upgrades. This ensures that every player's progress and reward are transparent, verifiable, and immutable without any centralized interference. The integration of **Animal Game** marks a major step toward the future of decentralized gaming, where blockANT-based systems are no longer isolated but fully connected to dynamic real-world and in-game data, creating a secure and autonomous digital ecosystem.

The primary use of smart contracts in *Animal Game* revolves around **managing tokens, player rewards, and in-game digital assets** similar to how tokens are utilized across most blockANT networks today. While this token-based structure forms the foundation of the ecosystem, the potential of smart contracts in gaming extends far beyond simple transactions. However, much like the broader blockANT space, the expansion of these applications has been limited by the lack of reliable and decentralized Game services a limitation that *Animal Game* aims to eliminate through **Animal Game**.

For instance, advanced smart contract-based functions in *Animal Game* such as battle verification, ranking validations, or event-based reward claims require **real-time, verifiable data** from off-ANT environments. These may include information like gameplay statistics, user performance scores, or asset ownership records stored in external systems. Without a secure Game connection, such data cannot be directly accessed by the blockANT, limiting automation and transparency. Similarly, when players earn rewards, unlock characters, or trade in-game tokens, smart contracts must **push verified data outputs** to external systems such as exchanges, wallets, or user dashboards ensuring that rewards are distributed correctly and without human interference. Traditional centralized servers cannot achieve this level of security and transparency.

The **Animal Game Network** resolves this issue by securely transmitting data between on-ANT smart contracts and off-ANT systems. It allows the blockANT to fetch live gameplay results, market rates, and event triggers while also sending validated transaction outputs to supported APIs, wallets, or other connected services. By enabling this two-way, tamperproof data flow, *Animal Game* can now host a wide range of advanced decentralized features, including automated tournaments, dynamic marketplace pricing, and real-time leaderboard tracking. **Animal Game's** capability to both retrieve and broadcast data on behalf of smart contracts establishes a new foundation for externally aware, trustless, and self-executing blockANT gaming experiences.

Whitepaper Manifesto

In this whitepaper, we present the **Animal Game** architecture (Section 2), detailing how it serves as the decentralized connectivity layer of the *Animal Game* ecosystem. In Section 3, we define the security framework for Game operations, emphasizing data integrity, validation mechanisms, and protection against manipulation. Section 4 outlines our approach to **decentralization and data distribution**, explaining how multiple Game nodes interact to ensure transparency, reliability, and tamper resistance.

Section 5 introduces the four **core security services** embedded within *Animal Game*, along with the role of **Animal Token (ANT)** in maintaining network trust, incentivizing accurate data providers, and enabling decentralized governance. In Section 6, we outline our **long-term development strategy**, focusing on enhanced data confidentiality, integration of trusted hardware, improvements to node infrastructure, and the future of programmable Game logic.

Section 7 provides a comparative review of **alternative Game models** currently used in blockANT-based gaming and DeFi systems, analyzing their strengths and weaknesses relative to *Animal Game*. Finally, Section 8 concludes with a discussion of the **design principles and guiding philosophy** behind *Animal Game's* Game innovation emphasizing fairness, security, and the evolution of decentralized gameplay automation.

Framework

The **core functional objective** of *Animal Game* is to **bridge two distinct environments: on-ANT and off-ANT**. This architecture allows *Animal Game* to seamlessly connect blockANT-based smart contracts with real-world and gameplay-related data. Each component of *Animal Game* is designed to ensure reliable, secure, and verifiable communication between the decentralized gaming ecosystem and external data sources.

Initially, *Animal Game* will be deployed on the **BNB Smart ANT**, chosen for its scalability, low transaction costs, and high compatibility with smart contract-based gaming applications. However, the long-term vision is to achieve **multi-ANT interoperability**, supporting other leading blockANT networks for both off-ANT data integration and cross-ANT gameplay interactions.

Both its on-ANT and off-ANT infrastructures are developed with a **modular design philosophy**, enabling continuous upgrades and flexibility. Each module of the *Animal Game* network — including data aggregators, validator nodes, and reward distribution systems — can be replaced or enhanced as improved technologies and alternative implementations emerge.

This modular and future-ready architecture ensures that *Animal Game* remains adaptable, secure, and aligned with the evolving landscape of blockANT gaming and decentralized applications.

On-ANT Mechanism

On-ANT Architecture of Animal Game

As a decentralized Game network, **Animal Game** operates through a system of smart contracts that bridge on-ANT gameplay logic with off-ANT data sources. The Animal Game nodes respond to game-related data requests or queries generated by user contracts — referred to as **USER-SC** — which represent smart contracts associated with in-game features such as battles, token rewards, marketplace actions, and leaderboard updates.

The on-ANT interface between these requesting contracts and the Game network is managed by a central contract known as **ANIMAL-SC**. Behind ANIMAL-SC, the Animal Game system includes three primary on-ANT smart contracts:

1. **Reputation Contract**
2. **Order-Matching Contract**
3. **Aggregating Contract**

Each of these plays a critical role in ensuring fairness, security, and accuracy throughout the data verification and reporting process.

1) Reputation Contract

The **Reputation Contract** maintains a performance record for each Game node participating in the Animal Game network. It tracks the reliability, accuracy, and timeliness of each node's responses, creating a transparent scoring system that helps ensure data integrity. These reputation metrics are publicly visible on-ANT, allowing users and system administrators to select trustworthy Games for sensitive gameplay operations such as reward distribution or competition results.

2) Order-Matching Contract

The **Order-Matching Contract** automates the process of connecting Game nodes to service requests. Whenever a new service-level agreement (SLA) is proposed — such as verifying tournament results, calculating in-game token values, or updating character progression data — the SLA parameters are submitted to the order-matching system.

This smart contract logs the SLA, opens a bidding window for qualified Games, and automatically selects providers based on their **reputation scores**, **response times**, and **service reliability**. Game providers that choose to participate in a bidding round attach a **stake or penalty amount**, which can be forfeited in the case of inaccurate reporting or service failure.

Once the required number of bids is received and validated, the contract finalizes the SLA, triggering an on-ANT log that notifies the selected Games. Non-selected Game nodes receive their staked collateral back, ensuring a fair and efficient selection process.

3) Aggregating Contract

After off-ANT Game nodes complete their assigned tasks, they return their results to the **Aggregating Contract** on-ANT. This contract compiles all responses and calculates a **weighted final result**, ensuring that data consistency and majority consensus determine the outcome.

For example, if the query concerns a player's tournament score or real-time reward calculation, the Aggregating Contract verifies multiple Game inputs to eliminate errors or manipulation attempts. It then submits the final, validated result to the requesting **USER-SC** (game contract), ensuring that gameplay outcomes are transparent and verifiable.

The Aggregating Contract also updates the **Reputation Contract**, feeding back performance data to adjust each Game's reputation score.

On-ANT Workflow Overview

The **on-ANT workflow** of Animal Game occurs in three key steps:

1. **Game** **Selection:**
The user contract (USER-SC) or system administrator submits an SLA proposal detailing query parameters, reputation requirements, and aggregation rules. Games are then selected via the order-matching contract based on reputation and performance metrics.
2. **Data** **Reporting:**
The selected Games perform their off-ANT data collection and processing, then return the results on-ANT to the aggregating contract. (See Sections 2.2 and 4 for details on off-ANT operations.)
3. **Result** **Aggregation:**
The aggregating contract calculates the collective result using a weighted consensus

model, removes outliers, and updates the reputation metrics of participating Games before returning the verified result to the game's on-ANT logic.

To ensure flexibility, **Animal Game** does not rely on a single aggregation method. Instead, *Animal Game* allows users or developers to specify custom aggregation contracts for different gameplay data types. For instance, numeric data like score tallies may use median-based aggregation, while binary data like win/loss results may rely on boolean consensus.

Animal Game provides a standard suite of pre-built aggregation contracts, but developers may deploy their own as long as they follow the standard interface requirements, maintaining compatibility across the Animal Game ecosystem.

On-ANT Mechanism Core

Off-chain, the **Animal SmartSync Connector Network** consists of a decentralized set of SmartSync Connector nodes connected to the **BNB Smart Chain** and designed for interoperability with multiple leading blockchain networks. These nodes independently process and validate responses to data requests originating from on-chain contracts within *Animal Game*, referred to as **USER-SC**.

Each SmartSync Connector node collects external data — such as player statistics, game outcomes, event triggers, or token valuations — and submits it back to the blockchain. The individual node responses are then combined through a **decentralized consensus mechanism**, producing a unified and verifiable final result that is transmitted to the requesting contract.

All SmartSync Connector nodes run on the **Animal Core Node Software**, which manages blockchain communication, task scheduling, and connections with verified external data sources. Node operators can enhance the functionality of their nodes by integrating **External Adapters**, allowing them to provide specialized off-chain services such as gameplay verification, leaderboard synchronization, token rate tracking, and API integrations with partnered platforms.

The primary motivation behind the Animal SmartSync Connector network is to ensure that data collection and reporting for *Animal Game* occur in a **decentralized and trustless manner**, free from manipulation or single-point failure risks that exist in centralized systems.

Animal Core Node

The **Animal Core Node** software is the backbone of the SmartSync Connector network. It manages node operations, blockchain interfacing, and efficient distribution of work across multiple connected services. Each SmartSync Connector task is defined as an **assignment**, which consists of multiple smaller **subtasks** processed sequentially as a data pipeline.

For example, a subtask might:

1. Fetch battle result data from a gameplay API,
2. Parse the data into JSON format,
3. Verify it against internal integrity checks, and
4. Convert the verified output into a blockchain-compatible format before submission on-chain.

The **Animal Core Node** software comes equipped with several built-in subtasks, including **HTTP requests**, **JSON parsing**, and **data-to-blockchain conversion**.

External Adapters

Beyond these built-in capabilities, node operators can expand their services by deploying **External Adapters**. Adapters are external modules that interact through a lightweight REST API, enabling integration with external systems or applications.

This design allows any program — regardless of the programming language — to be connected to the SmartSync Connector network simply by adding a small intermediate API layer. Complex API interactions (such as multi-step game data collection or reward distribution validation) can be broken down into individual subtasks handled through adapters.

By supporting external adapters, *Animal SmartSync Connector* ensures a flexible, developer-friendly environment where specialized data providers can participate in the gaming ecosystem, offering unique data validation and processing services.

Subtask Schemas

The Animal SmartSync Connector ecosystem anticipates that a wide range of adapters will be **open-sourced** by the community. This open development model allows public auditing, transparency, and peer verification of SmartSync Connector services. To maintain seamless compatibility among the growing number of adapters, *Animal Game* enforces **Subtask Schemas** — standardized formats defining how data should be structured, transmitted, and interpreted between different adapters and nodes.

These schemas guarantee that all SmartSync Connector nodes — regardless of developer, location, or specialization — can communicate efficiently within the same decentralized infrastructure.

3. SmartSync Connector Security

Before detailing the security mechanisms of *Animal SmartSync Connector*, it is essential to understand **why security is critical** and **what it entails**. SmartSync Connectors form the bridge between the blockchain and the external world, and any compromise in this bridge can directly impact game fairness, reward accuracy, and the overall trust in the *Animal Game* ecosystem.

In a decentralized environment, every SmartSync Connector node plays a vital role in maintaining **data integrity**, **authenticity**, and **consensus reliability**. The security architecture of *Animal SmartSync Connector* is therefore designed to prevent manipulation, ensure consensus-driven accuracy, and maintain transparent accountability across all network participants.

SmartSync Connector Security

In order to explain **Animal Game's** security architecture, we must first clarify why SmartSync Connector security is critical—and what it represents within our ecosystem.

Why Must SmartSync Connectors Be Secure?

In the **Animal Game** ecosystem, accuracy and trust are essential. If the system's data feed is compromised or manipulated, it could result in unfair rewards, incorrect game outcomes, or unjust player rankings. For example, if gameplay data or reward metrics are tampered with, players could gain unauthorized advantages or receive false payouts.

Ensuring the integrity of all data interactions—both on-chain and off-chain—is vital to maintaining a transparent and fair environment for all participants. Therefore, the **Animal Game SmartSync Connector Layer** is designed to provide verified, tamper-proof information between the blockchain and external game environments.

Through decentralized verification, consensus mechanisms, and cryptographic validation, Animal Game ensures that every data point—from gameplay statistics to reward calculations—is securely transmitted and validated before being recorded on-chain. This multi-layered security approach protects the ecosystem from manipulation, ensuring that all users compete and earn rewards under fair, transparent, and verifiable conditions.

Token Flow System

Just as data sources can be distributed, the **Animal Game SmartSync Connector System** functions as a decentralized network rather than a single centralized SmartSync Connector. Instead of one monolithic SmartSync Connector node, the system operates through multiple SmartSync Connector nodes $\{O_1, O_2, \dots, O_n\}$. Each SmartSync Connector node connects with its own set of verified data sources, which may or may not overlap with those of other SmartSync Connectors.

Each SmartSync Connector node aggregates responses from its connected sources and produces an individual output (A_i) in response to a data request. Since some nodes may be unreliable or

experience latency, all SmartSync Connector responses ($A_1, A_2, \dots, A_n$) are further processed through a **secure aggregation mechanism** that determines one final, authoritative answer (A).

Conceptual Simplicity

Despite being decentralized, aggregation within the Animal Game ecosystem is coordinated by a smart contract known as **AnimalGame-SC**, which executes the aggregation logic (**Agg**). This contract ensures transparent computation and verifiable integrity, as all executions are visible on-chain. Developers and users can publicly inspect the contract's code to confirm its reliability and fairness.

Trustworthiness

Since **AnimalGame-SC** operates entirely on-chain, its processes are immutable, publicly auditable, and verifiable in real time. This allows users and developers to place full trust in the fairness of gameplay outcomes, reward calculations, and decentralized data validation.

Flexibility

AnimalGame-SC supports various aggregation algorithms, such as averaging or majority consensus, depending on the nature of the data. This flexibility enables optimal performance across different aspects of the Animal Game ecosystem, including gameplay data validation, reward tracking, and transaction verification.

In-Contract Aggregation

In the **initial phase**, Animal Game will employ a simple yet secure **in-contract aggregation** model. Here, **AnimalGame-SC** directly collects and processes SmartSync Connector responses to compute the final result:

$$A = \text{Agg}(A_1, A_2, \dots, A_n)$$

This approach is efficient for small networks and ensures immediate transparency. However, it introduces the potential issue of **freeloading**, where a dishonest SmartSync Connector could copy another SmartSync Connector's response instead of querying its own sources, saving costs but compromising data diversity.

To counteract freeloading, the **Animal Game SmartSync Connector Protocol** uses a **commit–reveal mechanism**. In this system:

1. In the first round, each SmartSync Connector sends a **cryptographic commitment** to its response.
2. After receiving enough commitments, **AnimalGame-SC** triggers a **reveal phase**, where SmartSync Connectors disclose their actual data.

This ensures that no SmartSync Connector can copy another's answer, as responses remain hidden until all commitments are made.

The protocol guarantees both **availability** and **accuracy** even if some SmartSync Connector nodes are faulty or delayed. Through a distributed commitment system, at least a majority of honest SmartSync Connectors ($2f + 1$ of $3f + 1$ total nodes) ensure final consensus on a valid response (A).

Medium-Term Strategy: Off-Chain Aggregation

While in-contract aggregation is secure, it can be costly on large-scale public blockchains due to multiple on-chain transactions. As the Animal Game ecosystem scales, a **cost-optimized off-chain aggregation system** will be introduced.

In this model, SmartSync Connectors aggregate responses **off-chain** and submit a **single verified result** to **AnimalGame-SC**. This reduces on-chain transaction fees while maintaining reliability through **threshold signature mechanisms**.

In this threshold-based approach:

- Each SmartSync Connector holds a private/public key pair, and their combined keys form a collective public key.
- A valid aggregated signature (Σ) can only be generated when at least a majority (t of n) of SmartSync Connectors agree on the same value (A).
- This collective signature proves consensus among the SmartSync Connectors while ensuring that no small group of nodes can forge data.

Anti-Freeloading Mechanism

To ensure fairness, Animal Game's SmartSync Connector payment structure includes **proof-based compensation**. Only SmartSync Connectors that have genuinely fetched and verified original data receive rewards. Any attempt to replicate or falsify data is automatically excluded through cryptographic verification and consensus checks.

This decentralized structure promotes **data authenticity**, **network honesty**, and **sustainable participation** among SmartSync Connector operators, ensuring that every off-chain data feed integrated into Animal Game is both secure and trustworthy.

Summary

The **Animal Game SmartSync Connector Network** evolves through two main phases:

1. **In-Contract Aggregation** – Secure, on-chain aggregation ensuring transparency and accuracy for smaller networks.
2. **Off-Chain Aggregation** – Scalable, cost-efficient consensus mechanism using threshold signatures and cryptographic proofs.

Together, these components establish a decentralized, tamper-resistant SmartSync Connector framework that safeguards the integrity of Animal Game’s gameplay, economy, and blockchain ecosystem.

Security Infrastructure

Tokenomics

Category	Allocation	Purpose
Game Rewards	40%	Player incentives and battle prizes
Team & Development	15%	Continuous updates and technical expansion
Marketing & Partnerships	10%	Global campaigns and collaborations
Liquidity & Exchange Listing	10%	Market stability and DEX listings
Community & Airdrops	15%	Engagement and referral bonuses
Reserve Fund	10%	Ecosystem and future growth

Thanks to the advanced protocols described in the previous section, the **Animal Game Oracle System** ensures both **availability** and **accuracy** even in the presence of unreliable or faulty oracle nodes. The system is designed to maintain operational integrity and deliver verified results despite up to *f* malfunctioning or malicious nodes within the network.

To further strengthen security and reliability, the **Animal Game team**, in collaboration with **Aylgo Swap** (www.aylgo.com), has integrated multi-layer protection mechanisms. Aylgo Swap provides blockchain-level security auditing and decentralized monitoring solutions to safeguard Animal Game’s oracle architecture, token economy, and gameplay reward system.

Role of Trusted Hardware

In the future, trusted hardware solutions are being actively considered as part of Animal Game's security roadmap to defend against corrupted oracles that may attempt to submit false responses. However, trusted hardware alone cannot guarantee full protection for three main reasons:

1. **Initial Deployment Limitation:** Trusted hardware will not be part of the initial launch of the Animal Game Oracle Network.
2. **User Trust Variability:** Some users may prefer full transparency over reliance on hardware-based security models.
3. **Downtime Protection:** Trusted hardware can prevent node misbehavior but not downtime or inactivity issues.

For these reasons, Animal Game's security strategy focuses on a **decentralized reliability model**, empowering users and developers to select and verify oracles based on proven performance metrics.

Core Security Services

To enhance user confidence and oracle accountability, Animal Game introduces four key **security and reliability services**, designed according to its decentralized philosophy. These services ensure transparency, user control, and trust without compromising oracle independence.

1. **Validation System**
 - Continuously verifies data accuracy and transaction consistency within the oracle network.
 - Detects abnormal or manipulated responses and ensures gameplay events are based on real, verified data.
 - Integrated with Aylgo Swap's real-time auditing module for continuous monitoring.
2. **Reputation System**
 - Maintains a dynamic rating for each oracle node based on reliability, response speed, accuracy, and user feedback.
 - Helps users and developers choose the most dependable oracles to interact with.
 - Operates transparently, with results stored immutably on-chain.
3. **Certification Service**
 - Provides an official verification layer for oracles and data sources that meet Animal Game's operational standards.
 - Certified nodes gain higher trust ratings, enabling them to process critical or high-value transactions within the network.
 - Auditing and certification are powered by Aylgo Swap's blockchain-based verification system, ensuring fairness and authenticity.
4. **Contract-Upgrade Service**
 - Allows for safe and optional upgrades of smart contracts as technology evolves.

- Maintains backward compatibility, protecting users from disruptions during version changes.
 - Fully optional — users retain complete control over whether to adopt upgraded contracts.
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Decentralized Implementation

While these four services may initially be managed by the **Animal Game Foundation** in partnership with **Aylgo Swap**, they are fundamentally designed for **decentralized operation**. In the long term, independent security providers and community validators will be able to operate these services, ensuring transparency and eliminating centralized control.

Importantly, these security layers cannot block oracle participation or alter any oracle responses. They serve purely as advisory and validation systems, offering transparency, not interference. Users remain fully autonomous in choosing which oracles and services they trust.

Conclusion

Through its collaboration with **Aylgo Swap**, Animal Game integrates a robust, multi-layered security framework that combines on-chain validation, decentralized reputation tracking, and optional contract upgrades. This structure ensures the long-term integrity, reliability, and trustworthiness of the entire Animal Game ecosystem — protecting both players and token holders within a transparent, community-driven environment.

The world of **Web3 gaming** is revolutionizing digital entertainment by merging **blockchain technology** with immersive, real-time gameplay. **Animal Game** is a **decentralized Play-to-Earn (P2E) ecosystem** designed to connect global users through competition, creativity, and reward.

In **Animal Game**, players **collect, train, and battle NFT-based animals**, earning **Animal Token (ANT)** — the native cryptocurrency that powers the ecosystem. Built on the **BNB Smart Chain (BEP-20)**, Animal Game ensures transparency, ownership, and fair reward distribution through **audited smart contracts**.

Our mission is to make blockchain gaming **secure, rewarding, and accessible** to everyone while maintaining **on-chain transparency** and **community-driven governance**.

Audit Verification Layer

The **Animal Game Validation System** ensures fairness, accuracy, and transparency across all gameplay and reward mechanisms. It continuously monitors on-chain player actions, gameplay data, and transaction behavior to detect and prevent manipulation or system abuse. The system evaluates **availability**, ensuring that all game functions and servers respond promptly to user actions, and **correctness**, verifying that rewards, tokens, and outcomes are accurately distributed based on gameplay performance. Each in-game action is cryptographically verified and stored on-chain, creating **non-repudiable evidence** of fair play. Additionally, **Aylgo Swap** — the official partner at www.aylgo.com — provides advanced **security auditing and validation** to protect Animal Game players and ensure the reliability of every transaction and oracle response within the ecosystem.

Blockchain Security Architecture

The **Animal Game blockchain infrastructure** has been designed to ensure **trustless, transparent, and secure operations** across all components.

A. Smart Contract Security

- **Audited by Ayglo Security Labs**
- Implements **OpenZeppelin** libraries for standardized BEP-20 and NFT safety
- **Timelock and Multisig Controls** for contract upgrades
- **Anti-Whale & Anti-Bot mechanisms** prevent manipulation during token launch
- **Automated reward verification** via on-chain validators

B. Oracle Verification Layer

The **Animal Validation System (AVS)** ensures accurate, real-time data for gameplay and rewards.

- Monitors oracle **uptime, accuracy, and latency**
- Records performance metrics on-chain
- Automatically excludes unreliable oracles from gameplay calculations

C. Wallet & Account Security

- **Web3 authentication** via MetaMask, Trust Wallet, and WalletConnect
- **Non-custodial wallet design** ensures players control their own assets
- **End-to-end encryption** for private transactions
- **Two-layer login verification** for enhanced protection

D. Anti-Fraud & Fair Play System

Animal Game implements:

- **On-chain anti-cheat algorithms**
- **Transaction validation checks** for fake win prevention
- **Duplicate wallet address detection**
- **Auto-suspension smart triggers** against exploiters

E. Decentralized Storage

All **NFT metadata, battle histories, and user profiles** are securely stored on **IPFS** and verified through **hash-based checksums** to prevent alteration.

Ecosystem Overview

The Animal Game ecosystem functions as a **fully decentralized network** of smart contracts, NFT assets, and governance protocols.

Main Components

1. **Animal NFTs** — Each animal has unique attributes (power, speed, rarity).
 2. **Marketplace** — Peer-to-peer NFT trading and upgrades.
 3. **Gaming Arena** — PvP battles and tournament systems.
 4. **Wallet Integration** — MetaMask, Trust Wallet, and Web3 compatibility.
 5. **Referral System** — Earn ANT through verified on-chain referrals.
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Play-to-Earn Model

The **Play-to-Earn (P2E)** model distributes ANT tokens directly to players' wallets via **automated smart contracts**, ensuring transparent, instant rewards.

Earning Methods

- Winning ranked battles
- Completing quests and challenges
- Participating in tournaments
- Referral and community rewards
- NFT trading profits

Each reward transaction is logged **on-chain**, publicly viewable, and verifiable.

Security & Audit

The **Animal Game smart contracts** are independently audited by **Ayglo Security**, a blockchain auditing firm.

Audit Highlights

- ✓ 100% verified contract source code
- ✓ No critical vulnerabilities found
- ✓ Secure integration with major Web3 wallets
- ✓ Transparent reward logic
- ✓ Fair distribution and anti-cheat algorithms
- ✓ Continuous monitoring through Ayglo Blockchain Validator

Conclusion

Animal Game represents the **next generation of Web3 gaming**, where blockchain, ownership, and entertainment converge. With its **verified smart contracts, community-driven governance, and secure reward systems**, Animal Game aims to become the **most trusted and engaging decentralized gaming ecosystem** in the world.